

Supplementary Material S1

Extended Methods, Discussion, and AI-Assisted Development Workflow

Astri J. Lundervold

Birgitte Berentsen

Arvid Lundervold

Supplement to the manuscript “An AI-Assisted, Rule-Based Pipeline for Pre-Examination Screening in Clinical Neuropsychology: An Exploratory Method-Development Study” (Manuscript ID JINS-2026-0074).

This supplementary document provides the Extended Methods and Discussion that accompany the main manuscript, the AI-assisted development workflow, and a typeset rendering of the *prompt development material* for the deterministic rule-based pipeline, preserved in the project repository. It corresponds to the `manuscript/supplementary/rendered_prompt_source.tex`. The current manuscript also references separate attachment files (S2, S3) for the two illustrative case reports.

AI-Assisted Development Workflow

This section documents the AI-assisted development process itself, including the prompting strategy, the human-in-the-loop review, and model-card-style summaries of the assistant models. It complements—and does not replace—the verbatim prompt/system-prompt material rendered later in this document. We reiterate the governing principle stated in the manuscript: artificial intelligence was used *only during development*, never at runtime, and *no patient data were provided to any model at any point*; development used the published instrument specifications and synthetic or fully de-identified example records.

1. Prompting strategy: an iterative, clinician-led sequence

Development followed an iterative “design brief → draft → review → correct” loop, led by the clinical authors, rather than a single prompt. The main stages were as follows; the verbatim source prompts referenced here are reproduced in the “Rendered Project Material” section below.

- 1. Problem framing and design brief.** The clinical authors specified the intended use (pre-examination screening decision support), the five instruments and their published cutoffs, the three target audiences, and the required structured (JSON) output. This brief—not the model—defined every clinical rule and threshold.
- 2. Initial elicitation** (ORIGINAL_PROMPT_20260207, Claude Opus 4.6). An exploratory prompt was used to draft a first reasoning-and-reporting structure and to surface candidate domain logic. Outputs were treated as drafts only.
- 3. Specification and refactoring** (UPDATED_PROMPT_20260316 and SYSTEM_PROMPT_for_analysis v1.1, GPT-5.4). After the analysis-ready cohort file was prepared, the design brief was formalized into a system-prompt specification, and the assistant was asked to (re)implement the pipeline (`scripts/neuropsych_pipeline.py`) against that specification.

4. **Rule extraction and source verification.** Each AI-proposed classification rule, cutoff, and cross-domain trigger was checked line-by-line against the cited source instrument (BIS, CPT-3, Chalder, HADS, RBANS) and corrected where the draft diverged. Cutoffs were locked to published values; the model was not permitted to “choose” thresholds.
5. **Visualization and audience tailoring.** Prompts iterated on the multi-panel and radar figures and on the three audience variants (clinical peers, referring physicians, patients/families), with the authors adjusting labels, polarity (higher = better), and the color-blind-safe palette.
6. **Report assembly and reproducibility.** Final prompts addressed deterministic report generation (LaTeX/PDF), removal of any runtime model/network dependency, and packaging for reproducible execution (`environment.yml`).

A recurring pattern across stages was *correction*: the assistant’s drafts were frequently edited or rejected (e.g., tightening a cutoff to the published value, removing an over-confident causal claim, or re-labelling a “diagnosis” as a flagged finding). The final rules reflect author decisions, not model defaults.

2. Human-in-the-loop review and version control

The authors acted as human-in-the-loop reviewers throughout the design phase (principally February–March 2026; see the dated source prompts reproduced in the “Rendered Project Material” section below). Every AI-proposed rule, cutoff, narrative template, and figure was *accepted*, *corrected*, or *rejected* by an author before being incorporated. The public repository (<https://github.com/arvidl/AI-driven-neuropsych-screening>) records the released, reviewed state of this work rather than every intermediate iteration: the pipeline code was imported under version control by git. The preserved prompt and system-prompt material—with its dated original prompt (2026-02-07, Claude Opus 4.6) and updated prompt/system-prompt specification (2026-03-16, GPT-5.4)—documents the design briefs and their successive refinement and serves as the primary record of the AI-assisted development process.

Two invariants were maintained: (i) no real or identifiable patient data were ever entered into the assistant; and (ii) the deployed pipeline contains no model, no AI client library, and no network calls, so AI cannot influence any patient-level output at runtime. Verification of the deployed pipeline (quantitative accuracy, clinical coherence after data cleaning, and output completeness) was performed by the authors on the analysis-ready cohort file, as described in the manuscript.

3. Model cards (development-time assistants)

The following concise, clinically oriented summaries describe the assistant models as *used in this project*. They are not a substitute for the providers’ official documentation; adopters should consult the current official model cards, because model versions, capabilities, and terms change over time. The versions, release dates, and access terms below were checked against the providers’ official cards and Cursor’s data-governance documentation (accessed June 2026; full sources listed after the cards).

Model card A — Anthropic Claude Opus 4.6

Provider / version	Anthropic; Claude Opus 4.6 (accessed via the Cursor IDE).
Access mode	Enterprise configuration with code-/prompt-privacy and data-retention controls enabled (training on submitted content disabled; see source [c]).
Role in project	Development-time coding/design assistant: initial elicitation of the reasoning-and-reporting structure and candidate domain logic.
Inputs provided	Published instrument specifications; design briefs; synthetic or fully de-identified example records; project source code.
Inputs <i>not</i> provided	No real or identifiable patient data; no cohort records transmitted to the provider.
Determinism	Generative and non-deterministic; all outputs treated as drafts and verified by the authors.
Intended use here	Drafting code, rules, templates, and figures for subsequent human verification.
Known limitations	May produce fluent but unsupported statements (hallucination); not a medical device; not validated for autonomous clinical use; outputs require expert review.
Runtime role	None (no model is present in the deployed pipeline).
Official documentation	Anthropic, <i>Claude Opus 4.6 System Card</i> (released 5 February 2026; API identifier <code>claude-opus-4-6</code>); see source [a].

Model card B — OpenAI GPT-5.4

Provider / version	OpenAI; GPT-5.4 (accessed via the Cursor IDE).
Access mode	Enterprise configuration with code-/prompt-privacy and data-retention controls enabled (training on submitted content disabled; see source [c]).
Role in project	Development-time coding/design assistant: formalizing the system prompt and (re)implementing the pipeline against the specification.
Inputs provided	Published instrument specifications; the formal system-prompt specification; synthetic or fully de-identified example records; project source code.
Inputs <i>not</i> provided	No real or identifiable patient data; no cohort records transmitted to the provider.
Determinism	Generative and non-deterministic; all outputs treated as drafts and verified by the authors.
Intended use here	Implementing scoring logic, cross-domain rules, visualizations, and report assembly for subsequent human verification.
Known limitations	May produce fluent but unsupported statements (hallucination); not a medical device; not validated for autonomous clinical use; outputs require expert review.
Runtime role	None (no model is present in the deployed pipeline).
Official documentation	OpenAI, <i>GPT-5.4 Thinking System Card</i> (5 March 2026; API identifier <code>gpt-5.4</code>); see source [b].

Provider documentation (accessed June 2026).
[a] Anthropic, “Claude Opus 4.6” and the accompanying Claude Opus 4.6 system card, <https://www.anthropic.com/news/claude-opus-4-6>.
[b] OpenAI, “GPT-5.4 Thinking System Card,” <https://openai.com/index/gpt-5-4-thinking-system-card/>.
[c] Cursor, “Privacy and Data Governance,” <https://cursor.com/docs/enterprise/privacy-and-data-governance> (“With Privacy Mode enabled your code is never used for training by Cursor or other AI model providers”; Privacy Mode is on by default for Enterprise teams, under zero-data-retention agreements with model providers).

Extended Methods and Discussion

To keep the main text within the journal word limit, several points that were elaborated at length are summarised in the manuscript and given in full here. Each subsection below is referenced from the corresponding place in the main text.

E1. Rule-based versus generative-AI approaches, and the actuarial tradition

Automated, rule-based scoring and narrative interpretation of psychological and neuropsychological tests is not new: commercial computer-based test interpretation systems for instruments such as the MMPI and the Wechsler scales have existed for decades, and their validity and limitations have

been debated at length (Butcher et al., 2000). Such systems are best understood as actuarial—they apply fixed cutoffs and pre-authored interpretive text—and a large body of evidence indicates that mechanical, actuarial combination of data is often as accurate as, or more accurate than, unaided clinical judgment (Dawes et al., 1989; Grove et al., 2000). The contribution of the present work therefore does not lie in the existence of rule-based scoring as such. It lies in two places: first, in the *development workflow*—whether a practising clinician can use a general-purpose AI coding assistant to specify, implement, and audit a transparent multi-domain screening tool; and second, in the integration step—structuring an explicit cross-domain pass and audience-tailored communication on top of single-instrument scoring.

This also clarifies how a rule-based approach differs from a generative-AI (large language model, LLM) approach to the same task. An LLM-based assistant could in principle ingest a free-text referral letter and unstructured history and produce a fluent narrative synthesis, which is a genuine advantage when the input is heterogeneous prose. The costs are equally concrete: LLM outputs are non-deterministic, can be difficult to audit, may fabricate plausible-sounding but unsupported statements (Ji et al., 2023), and—if invoked at the point of care—raise data-governance and privacy questions because patient data must be transmitted to a model (DeRight & Puente, 2026; Meskó & Topol, 2023). A rule-based system makes the opposite trade-off: it requires its inputs to be supplied as structured, validated variables rather than free text, and it cannot generalize beyond its encoded rules, but in return every output is deterministic, reproducible, inspectable, and free of runtime data transmission. The framework presented here deliberately occupies this second position, and uses AI only during development.

E2. Rationale for confining AI to development

A natural question is why an AI coding assistant should be used at all when the resulting runtime logic could have been written by hand. We make no claim that the AI produces an algorithm that hand-coding could not. The rationale is twofold and is deliberately separated from runtime behaviour. (1) *Lowering the barrier to building auditable tools*. Confining AI to the design phase—where its flexibility in drafting code, decision rules, narrative templates, and visualizations is most useful—and excluding it from the point of care—where determinism, reproducibility, privacy, and auditability matter most—lets a clinician-led team specify and implement a non-trivial multi-domain tool (scoring logic, cross-domain rules, three audiences of visual output, and a reproducible report build) in a fraction of the time and specialist programming effort that bespoke development would traditionally require. This is precisely the feasibility question this exploratory study examines. (2) *Safety and governance*. Because the final artefact contains no model, the patient-facing system inherits none of the non-determinism, hallucination risk (Ji et al., 2023), or data-transmission requirements of a generative system; every output is traceable to an explicit, inspectable rule. The trade-off is that the runtime tool is only as good as the rules a clinician chooses to encode and cannot adapt beyond them.

E3. RBANS: normed indices versus cohort-standardized subtests

We use the RBANS in two different ways. For the cohort-level group comparison in Table 1 of the main text we report the conventional age-corrected RBANS *index* scores, which are population-normed. For the patient-level pipeline we deliberately use *cohort-standardized raw subtests* rather than the normed indices. The rationale is transparency and self-containment: the open pipeline encodes only rules and cutoffs that we can publish, whereas the RBANS normative conversion tables are proprietary and cannot be redistributed; cohort-relative standardization keeps the demonstration

fully reproducible from the shared code and makes the patient-level percentiles explicitly cohort-referenced. The cost is that these percentiles are not population-normed and are not interchangeable with clinical RBANS index scores; we list this as a limitation, and in a normed deployment the same flag logic would be applied to validated index/percentile inputs.

E4. Data governance, privacy, and safety

The governing principle is that artificial intelligence touches the project only during development and never touches patient data. (i) *Development phase*. The AI coding assistant (Cursor IDE with Claude Opus 4.6 and GPT-5.4) was used to draft code, rules, templates, and figures from published instrument specifications and synthetic or fully de-identified example records; no real patient data, and no data derived from identifiable participants, were entered into the assistant or transmitted to any model provider. The assistant was used under an enterprise configuration with code-/prompt-privacy and data-retention controls enabled (training on submitted content disabled); the precise contractual limits of such enterprise subscriptions vary by provider and over time, and adopters should verify current terms rather than rely on defaults. (ii) *Runtime phase*. The deployed pipeline performs all computation locally, contains no model and no network calls, and therefore transmits no patient data anywhere; the analysis-ready cohort file was processed on institutional research infrastructure. (iii) *Standards consulted*. Data processing followed the study’s ethics approval (REK2015-01621) and the requirements of the EU General Data Protection Regulation (European Parliament and Council of the European Union, 2016) as implemented in Norwegian law, with reference to Norwegian Data Protection Authority (Datatilsynet) guidance on AI and personal data; we additionally used the risk-tiering logic of the EU Artificial Intelligence Act (European Parliament and Council of the European Union, 2024) to reason about where such a tool would sit (clinician-facing decision support under professional oversight, not autonomous diagnosis). These references are a transparent starting point for compliant local replication rather than formal legal advice.

E5. Missing-data handling

Missingness in this cohort is predominantly *blockwise* (structural) rather than scattered at the item level: where data are absent, it is typically because an entire instrument was not administered or not completed for a given participant (for example, the CPT-3 and RBANS require in-clinic administration, and a subset of participants lack one or more whole instruments), reflecting administration logistics rather than the participant’s responses to individual items. Within an administered instrument, item-level missingness was rare. We therefore did not impute and instead analysed each instrument on a complete-case basis, with the effective denominator reported for every measure. Because the dominant mechanism is instrument-level non-administration that is plausibly unrelated to the unobserved scores themselves, treating the data as missing at random (MAR) within domain is a reasonable working assumption (Schafer & Graham, 2002); a strict missing-completely-at-random (MCAR) assumption is not required for complete-case domain summaries and, given that instrument availability is tied to visit logistics, is unlikely to hold exactly. We cannot fully exclude a missing-not-at-random (MNAR) component (e.g., more symptomatic participants declining a session), and we therefore treat all cohort-relative percentiles and group comparisons as descriptive (Little, 1988).

E6. Screeners, differential diagnosis, anchoring, and a “challenge” mode

Two cautions about the inputs deserve emphasis. First, the instruments feeding the pipeline are screeners, not diagnostic instruments. Elevated scores on broad anxiety, fatigue, or mood screeners are non-specific and can reflect conditions the screener was never designed to identify—post-traumatic stress, panic disorder, or primary sleep pathology, among others—so a high screener score must not be read as a diagnosis, and differential diagnosis remains entirely a clinical task performed during and after the examination. Second, any tool that confidently presents an integrated summary carries a real risk of *anchoring* the clinician: a fluent, well-organized output can prime premature closure and amplify confirmation bias rather than counter it. This is a genuine limitation of the current design, which is oriented toward synthesizing and confirming the encoded findings. A constructive direction—arguably a more valuable role for automation than confirmation—is to invert this emphasis so that the system also surfaces what is *missing or underweighted*: explicitly listing differential diagnoses that the available screeners cannot rule out, flagging internal inconsistencies, and prompting the clinician to consider data the pipeline did not receive. Designed this way, the tool would challenge rather than reinforce the clinician’s initial impression, while leaving accountability for the formulation firmly with the clinician.

E7. Extended clinician interpretation of the two illustrative cases

The main text already gives the clinician’s interpretation of each case. To avoid duplication, this section does not restate that reasoning; it records only the additional points that did not fit the main-text word limit.

Case 1 (circumscribed sleep disturbance). The main-text interpretation (hypervigilance potentially lowering Detectability independently of sleep, and the need to separate sleep-mediated from learning-specific memory weakness) is extended here by the pre-examination planning it implies. The convergence of moderate insomnia, borderline attentional inefficiency, and verbal-learning weakness is most parsimoniously read as sleep disturbance acting as a principal upstream contributor, so sleep-focused intervention is a reasonable initial priority—but one to be held tentatively until the examination has disentangled sleep disruption from hypervigilance-driven arousal. Sleep-focused intervention should not, on its own, justify deferring the broader neuropsychological work-up.

Case 2 (multi-domain involvement). Beyond the main-text observations (a severe Naming deficit pointing to a language-network rather than a sleep/fatigue mechanism, and immediate rather than deferred follow-up), the radar and multi-panel figures make the relevant dissociation quantitatively explicit: near-average attention (about the 50th cohort percentile) against severely compromised memory and thinking (about the 3rd percentile). A separation of this magnitude is difficult to reconcile with a unitary sleep–fatigue account and is the specific feature that moves comprehensive neuropsychological assessment from a deferred to an immediate priority.

References (Extended Methods and Discussion)

- Butcher, J. N., Perry, J. N., & Atlis, M. M. (2000). Validity and utility of computer-based test interpretation. *Psychological Assessment*, 12(1), 6–18. <https://doi.org/10.1037/1040-3590.12.1.6>
- Dawes, R. M., Faust, D., & Meehl, P. E. (1989). Clinical versus actuarial judgment. *Science*, 243(4899), 1668–1674. <https://doi.org/10.1126/science.2648573>

- DeRight, J., & Puente, A. N. (2026). Artificial intelligence in neuropsychological practice: Tool selection, privacy, and billing guidance. *The Clinical Neuropsychologist*, 1–19. <https://doi.org/10.1080/13854046.2026.2635117>
- European Parliament and Council of the European Union. (2016). Regulation (eu) 2016/679 (general data protection regulation) [<https://eur-lex.europa.eu/eli/reg/2016/679/oj>].
- European Parliament and Council of the European Union. (2024). Regulation (eu) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act) [<https://eur-lex.europa.eu/eli/reg/2024/1689/oj>].
- Grove, W. M., Zald, D. H., Lebow, B. S., Snitz, B. E., & Nelson, C. (2000). Clinical versus mechanical prediction: A meta-analysis. *Psychological Assessment*, 12(1), 19–30. <https://doi.org/10.1037/1040-3590.12.1.19>
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 1–38. <https://doi.org/10.1145/3571730>
- Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83(404), 1198–1202. <https://doi.org/10.1080/01621459.1988.10478722>
- Meskó, B., & Topol, E. J. (2023). The imperative for regulatory oversight of large language models (or generative AI) in healthcare. *npj Digital Medicine*, 6(1), 120. <https://doi.org/10.1038/s41746-023-00873-0>
- Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, 7(2), 147–177. <https://doi.org/10.1037/1082-989X.7.2.147>

Rendered Project Material

For documentation purposes, the original verbatim supplement source has been preserved as `rendered_prompt_source.md`. The present file renders the same project material in a more manuscript-like layout while preserving the original structure, headings, lists, tables, and code blocks.

AI-driven Neuropsychological screening and report design

resulting in `scripts/neuropsych_pipeline.py` (2026-03-17 Cursor 2.6.20 with GPT-5.4)

ORIGINAL_PROMPT_20260207_Opus_46.md

PROMPT: Can you improve and refine this to be a better prompt: You are an AI-based clinical neuropsychologist, exposed to a cohort of 105 subjects - some with IBS, some without IBS (cfr. BGA_merged_all_20260208.csv). You are an expert in neuropsychological reasoning and interpretation of neuropsychological test profiles in multiple domains: cognition, emotion, sleep, distress, etc. (cf. BIS, CPT, FSS / Chalder, HADS, RBANS) related to patient demographics (e.g., Gender, Age, Education). The information you extract for each patient is actionable in terms of therapy advice, and the need or suggestion of further examinations. You are able to see the patient in context (cf. the cohort of 105 subjects). You also provide your output in JSON format containing information about findings within the different domains, your reasoning steps, and your conclusions. You also provide a didactic diagram with domain-specific profiles (cfr. BIS, CPT, FSS, HADS, RBANS) explaining the findings and empirical conclusions to your peers, to medical doctors, and to the patient and their closest persons.

UPDATED_PROMPT_20260316_GPT_54.md

PROMPT: I have now made an updated csv file (data/BGA_merged_all_20260208_cleaned_for_analysis.csv) and an updated Markdown file for system prompt (doc/SYSTEM_PROMPT_for_analysis.md) for updating our automated pipeline (neuropsych_pipeline.py) used to generate comprehensive, audience-tailored neuropsychological reports from multi-domain test data. Can you suggest the updated Python pipeline in scripts/neuropsych_pipeline_20260316.py accordingly?

SYSTEM_PROMPT_for_analysis.md (20260316 - GPT-5.4)

System Prompt for analysis (GPT-5.4): AI Clinical Neuropsychologist

Version: 1.1 **Date:** 2026-03-16 **Project:** AI-driven-neuropsych

You are an AI-based clinical neuropsychologist specializing in the interpretation of neuropsychological test profiles. You have access to a cohort dataset (BGA_merged_all_20260208_cleaned_for_analysis.csv) comprising 105 subjects, including both IBS patients and healthy controls.

Your expertise spans the following assessment domains and instruments:

- **Sleep / Insomnia:** Bergen Insomnia Scale (BIS)
- **Sustained attention / Executive function:** Continuous Performance Test (CPT)
- **Fatigue:** Chalder Fatigue Scale (11 items, bimodal scoring 0--11)
- **Emotional distress:** Hospital Anxiety and Depression Scale (HADS)
- **Neurocognitive function:** Repeatable Battery for the Assessment of Neuropsychological Status (RBANS)

Contextual variables: Gender, Age, Education, and IBS diagnosis status.

Tasks**1. Profile interpretation**

Analyze the individual's scores across all domains, identifying clinically meaningful patterns, elevations, and deficits relative to normative data and the cohort distribution.

2. Contextual reasoning

Situate the individual's profile within the full cohort (N=105), noting where they fall relative to group-level patterns (IBS vs. controls) and relevant demographic subgroups.

3. Clinical reasoning chain

Articulate step-by-step reasoning linking observed test patterns to plausible neuropsychological mechanisms (e.g., insomnia-fatigue-attention cascades, anxiety-sleep-cognitive performance interactions, gut-brain axis mediated effects on cognition and affect).

4. Actionable recommendations

Provide (a) therapy-relevant advice grounded in the profile, and (b) suggestions for further examinations or referrals where indicated.

5. Patient-specific graphical output

Generate didactic domain-profile visualizations using **Python with seaborn** (and matplotlib as needed). The output comprises two complementary figure types: a **multi-panel domain-profile figure** and a **radar/spider plot summary**. Both must serve three audiences and incorporate cohort context.

Visualization A: Multi-panel domain-profile figure

Implementation: Python, using **seaborn** and **matplotlib**. Save output as publication-quality PDF and screen-resolution PNG.

Core design:

For each of the five domains (BIS, CPT, Chalder, HADS, RBANS), produce a panel that shows:

- **Cohort context layer:**
 - Distribution of the full cohort (N=105) as a violin plot, strip/swarm plot, or KDE, split or color-coded by IBS status.
 - Normative/clinical cutoff lines where applicable (e.g., HADS ≥ 8 for caseness, BIS clinical threshold).
 - Shaded zones indicating normal, borderline, and clinical ranges.
- **Individual patient layer (overlaid):**
 - The patient's score highlighted as a prominent marker (e.g., large diamond or star) with annotation.
 - Percentile rank within the cohort and, where available, relative to published norms.
- **Subscale breakdown** where relevant:
 - HADS: separate anxiety (HADS-A) and depression (HADS-D) subscales.
 - RBANS: raw scores of all 12 subtests (Immediate Memory, Visuospatial/Constructional, Language, Attention, Delayed Memory, and Recognition).
 - CPT: performance metrics across all nine variables (omissions, commissions, reaction time, d', HRT, block change, variability, etc.).

Visualization B: Radar / spider plot summary

Purpose: Provide an at-a-glance holistic profile of the patient across all domains simultaneously. Especially suited for the **patient-facing summary** and for **physician quick-review**, where seeing the overall shape of strengths and difficulties matters more than exact numerical detail.

Implementation: Python, using `matplotlib` polar axes with optional `seaborn` styling context.

Design specification:

- **Axes:** One radial axis per domain, arranged as spokes. Use between 5 and 10 spokes depending on granularity:
 - **Compact version (5 spokes):** BIS (Sleep), CPT (Attention), Chalder (Fatigue), HADS (Mood), RBANS (Cognition).
 - **Expanded version (up to 10 spokes):** Breaks out subscales — BIS total, CPT d', CPT reaction time, Chalder total, HADS-A, HADS-D, RBANS Immediate Memory, RBANS Attention, RBANS Delayed Memory, RBANS Sum Raw.
- **Score normalization:** All domain scores must be transformed to a **common 0-100 percentile scale** (within the cohort) to allow meaningful cross-domain comparison on a shared radial range. Ensure consistent polarity: higher values = better functioning on all axes. Invert scales where raw high scores indicate worse outcomes (e.g., BIS, Chalder, HADS, CPT: high raw -> low percentile-of-wellness).

- **Layered overlays:**

Layer	Visual encoding	Purpose
Normative / healthy zone	Filled polygon in soft green (alpha ~ 0.15), bounded by cohort control-group median +/- 1 SD	Shows "typical healthy range"
IBS cohort median	Dashed polygon outline in muted orange	Shows typical IBS-group profile shape
Clinical concern zone	Shaded ring near center (low percentile) in soft red (alpha ~ 0.10)	Highlights areas falling below clinical thresholds
Individual patient	Bold solid polygon in saturated blue or teal, with vertex markers	The patient's own profile

- **Annotations on the patient polygon:**
 - Label each vertex with the patient's percentile value.
 - Flag vertices falling in the clinical concern zone with a small warning icon or color-shifted marker.
 - Add brief plain-language descriptors at each axis tip (e.g., "Sleep quality", "Concentration", "Energy", "Mood", "Memory & thinking").
- **Legend and interpretation guide:**
 - Include a compact legend identifying each polygon layer.
 - Add a one-sentence interpretation beneath the plot: e.g., *"Your profile shows relative strengths in memory and language, with areas of concern in sleep and fatigue that may be affecting your concentration."*

Audience-tailored view summary (both figure types)

Audience	Multi-panel figure	Radar plot
Clinical peers	Full statistical detail: z-scores, percentile ranks, confidence intervals, effect sizes relative to IBS/control subgroups. Technical axis labels and instrument nomenclature.	Expanded 10-spoke version. z-score and percentile annotations. Secondary inset with Cohen's d effect sizes per domain.
Referring physicians	Simplified summary panel: traffic-light color coding (green/amber/red) per domain, brief textual annotations highlighting clinical action points. Minimal jargon.	Compact 5-spoke version. Traffic-light vertex coloring. Flagged domains annotated with brief action notes.
Patient and close persons	Accessible infographic style: intuitive labels, plain-language annotations, larger fonts, explanatory legend. Avoid raw scores; use visual metaphors (e.g., gauge-style or horizontal bar with "your score" pointer against a "typical range").	Compact 5-spoke version. Qualitative anchors instead of numeric ticks. Plain-language interpretation sentence. Warm, non-alarming palette.

Output format — structured JSON

```
{
  "patient_id": "...",
  "demographics": { "age": "...", "gender": "...", "education": "...", "ibs_status":
    "..." },
  "domain_findings": {
    "sleep_BIS": {
      "scores": {},
      "percentile_in_cohort": "...",
      "interpretation": "...",
      "flags": []
    },
    "attention_CPT": {
      "scores": {},
      "percentile_in_cohort": "...",
      "interpretation": "...",
      "flags": []
    },
    "fatigue_Chaldler": {
      "scores": {},
      "percentile_in_cohort": "...",
      "interpretation": "...",
      "flags": []
    },
    "emotional_distress_HADS": {
      "scores": { "HADS_A": "...", "HADS_D": "...", "HADS_total": "..." },
      "percentile_in_cohort": "...",

```

```

    "interpretation": "...",
    "flags": []
  },
  "neurocognition_RBANS": {
    "scores": {
      "immediate_memory": "...", "visuospatial": "...",
      "language": "...", "attention": "...", "delayed_memory": "...",
      "total_scale": "..."
    },
    "percentile_in_cohort": "...",
    "interpretation": "...",
    "flags": []
  }
},
"cross_domain_patterns": "...",
"cohort_context": "...",
"reasoning_chain": ["Step 1: ...", "Step 2: ...", "..."],
"recommendations": {
  "therapeutic": [...],
  "further_examinations": [...]
},
"visualizations_generated": {
  "multi_panel": {
    "clinical": "{patient_id}_multipanel_clinical.pdf",
    "physician": "{patient_id}_multipanel_physician.pdf",
    "patient": "{patient_id}_multipanel_patient.pdf"
  },
  "radar_plot": {
    "clinical": "{patient_id}_radar_clinical.pdf",
    "physician": "{patient_id}_radar_physician.pdf",
    "patient": "{patient_id}_radar_patient.pdf"
  }
}
}

```

Guiding principles

- Always ground interpretations in established neuropsychological norms and clinical evidence.
- Distinguish between statistically notable and clinically significant findings.
- Be transparent about uncertainty and the limitations of cross-sectional cohort data.
- Tailor language complexity and visual detail to each target audience.
- Ensure all visualizations are colorblind-safe and accessible.
- In the radar plot, always normalize to a common percentile-of-wellness scale with consistent polarity (higher = better) to prevent misinterpretation across domains with different raw-score directionality.